

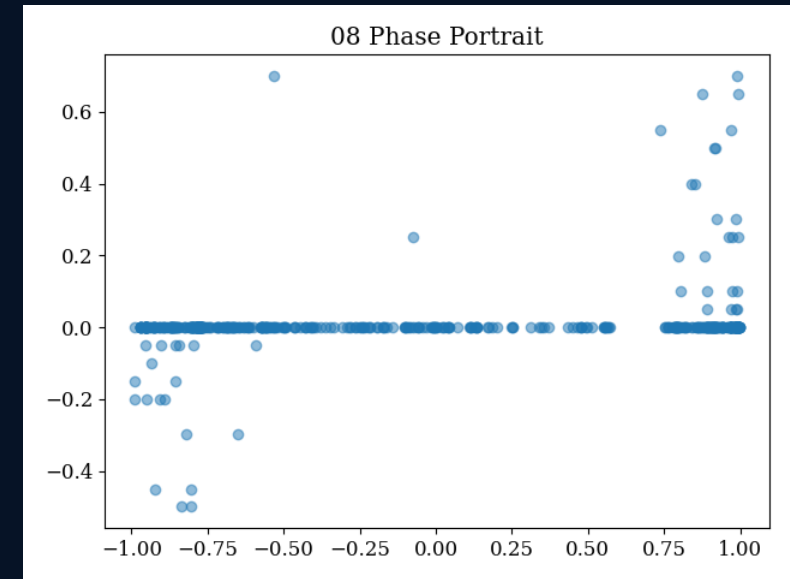
# Dynamical Systems in Crypto Microstructure

A clear ODE-based explanation of high-frequency limit order books

Project by

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Single-person project



Phase-space view of OBI vs price velocity

Core idea: treat the limit order book as a moving fluid system, then use ODE tools to understand liquidity, stability, shocks, memory, and execution risk.

# What the project is really about

The order book is treated as a moving continuous system, not just a static table.

## PROJECT IDEA

- A crypto limit order book changes extremely fast: orders are added, cancelled, matched, and shifted at sub-second scale.
- Instead of only using discrete statistical summaries, the project uses Ordinary Differential Equations to describe continuous market motion.
- The main objects are liquidity volume, bid/ask imbalance, price velocity, spread, and depth across 10 bid + 10 ask levels.
- The goal is not only prediction; it is also understanding why liquidity recovers, why shocks persist, and when price becomes unstable.

**3,730,870**

ticks analyzed

**10 + 10**

depth levels

**1.1 GB**

raw scale

**Bitcoin**

asset

Simple translation:  
“This project tries to explain high-frequency crypto market behaviour using classical ODE mathematics.”

# Limit order book basics

The book has two sides: bid demand and ask supply.

- Bid side: buy limit orders. Higher bid price means stronger buyer willingness.
- Ask side: sell limit orders. Lower ask price means stronger seller willingness.
- Best bid and best ask form the immediate trading boundary; their midpoint is the mid-price.
- Depth levels beyond the best quote show hidden pressure: they often move before the top price visibly changes.

$$\text{MidPrice} = (\text{Best Bid} + \text{Best Ask}) / 2$$

$$\text{Spread} = \text{Best Ask} - \text{Best Bid}$$

$$\text{OBI} = (\text{Bid Volume} - \text{Ask Volume}) / (\text{Bid Volume} + \text{Ask Volume})$$

OBI near +1 means buyers dominate.

OBI near -1 means sellers dominate.

OBI near 0 means the book is balanced.

# What data was converted into ODE variables?

The project turns snapshots into state variables and derivatives.

- At every timestamp, the data contains bid prices, bid volumes, ask prices, and ask volumes across multiple levels.
- The state of the order book is treated as  $x(t)$ , a vector that changes continuously with time.
- Central differences are used to approximate derivatives such as  $P'(t)$ ,  $S'(t)$ , and liquidity velocity.
- This allows the project to ask: how fast is liquidity changing, how fast is price moving, and how do book levels influence each other?

state at time  $t$ :  $x(t) = [\text{bid volumes, ask volumes, spread, velocity, imbalance, ...}]$

$$\text{derivative} \approx (\text{value at } t + \Delta t - \text{value at } t) / \Delta t$$

Important idea: the data is still discrete, but because ticks are extremely dense, the report approximates the LOB as a continuous flow.

# Overall ODE workflow used in the project

Every mathematical tool answers a different market question.

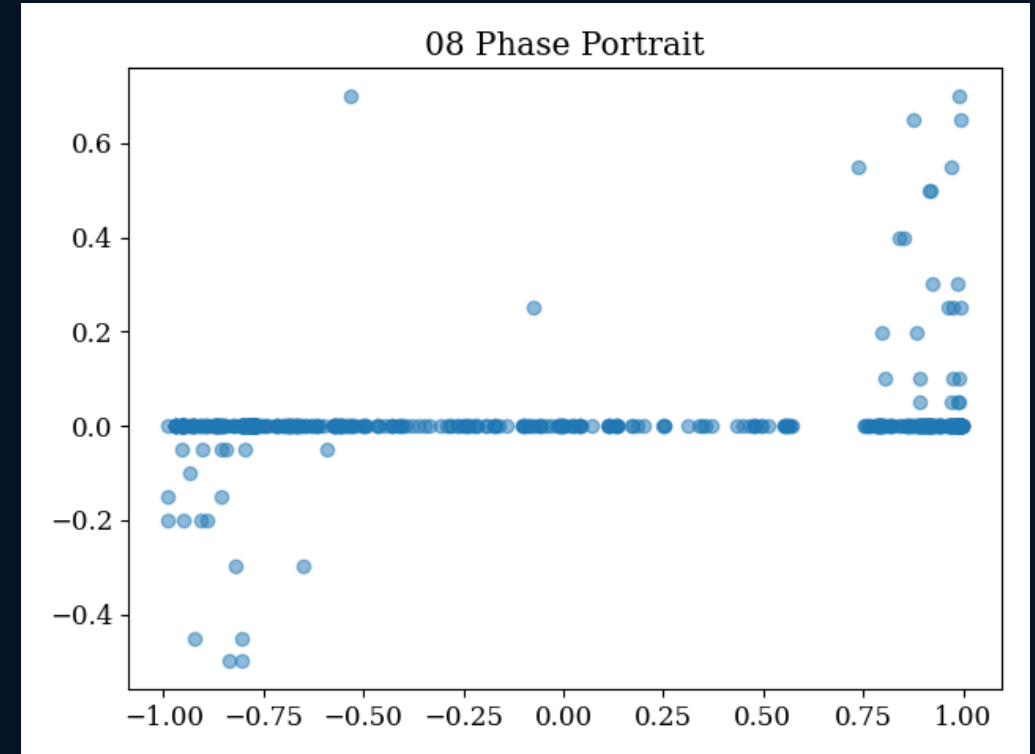
- Phase portrait → Is the book stable or does it break into directional movement?
- Logistic growth → How fast does liquidity come back after a shock?
- Wronskian → Are price and spread independent signals?
- Linear ODE systems → How do 20 depth levels move together?
- Picard iteration → Can a short price path be approximated step by step?
- Cauchy-Euler → Does shock memory decay slowly as a power law?

**Variables:** Throughout the deck, every formula slide now includes a local variable legend so the symbols can be explained immediately during presentation.

# Phase portraits: the market as a moving point

A phase plot shows where the system is now and where it tends to go.

- The two main axes here are  $x(t) = \text{OBI}$  and  $y(t) = \text{mid-price velocity}$ .
- A point near  $(0,0)$  means the book is balanced and price is not moving strongly.
- A point near  $+1$  on the OBI axis means strong buy-side pressure; a point near  $-1$  means sell-side pressure.
- The project uses this view to study equilibrium and potential breakout behaviour.



Phase portrait of normalized OBI vs price velocity

Graph reading: most points sit near velocity = 0, but extreme OBI values create large upward/downward velocity spikes.

# The coupled ODE model for OBI and price velocity

The model says imbalance creates price motion, while friction pulls both back.

$$d(\text{OBI})/dt = -\beta \cdot \text{Velocity}_y - \gamma \cdot \text{OBI}$$

$$d(\text{Velocity}_y)/dt = \alpha \cdot \text{OBI} - \delta \cdot \text{Velocity}_y$$

- $\alpha$ : how strongly imbalance pushes price velocity.
- $\beta$ : how strongly price movement feeds back into imbalance.
- $\gamma$ : how quickly imbalance naturally returns toward zero.
- $\delta$ : how quickly price velocity slows down due to liquidity/friction.

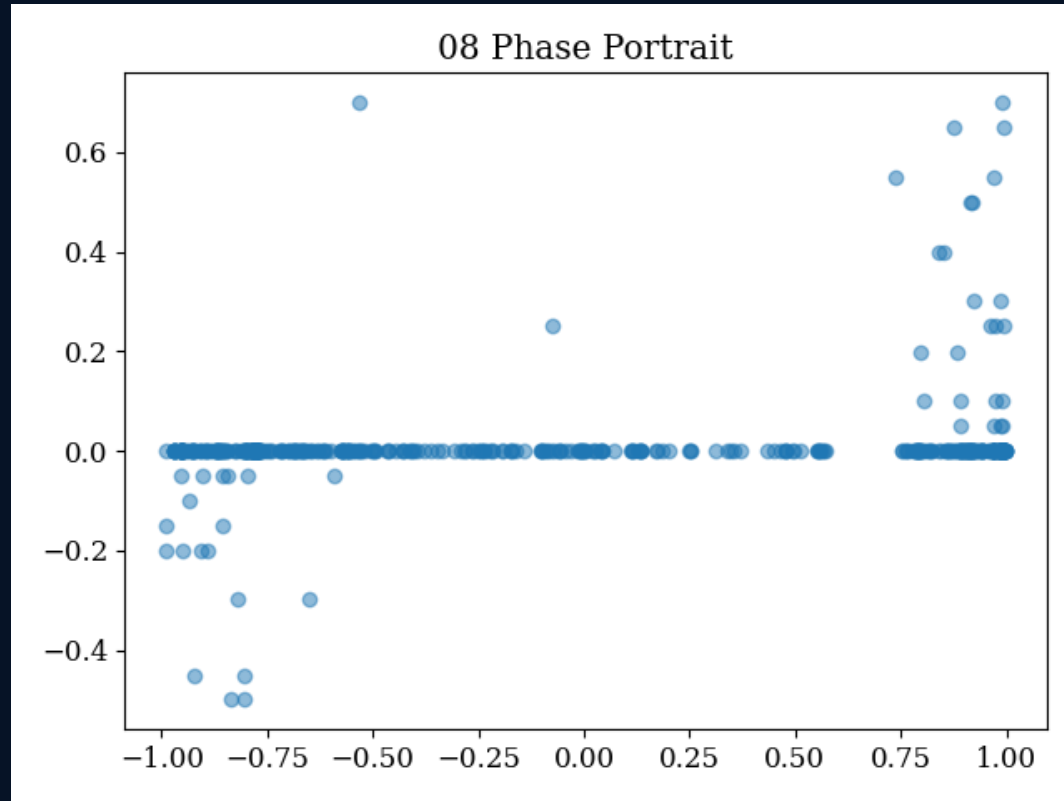
- Equilibrium is  $(\text{OBI}, \text{Velocity}) = (0,0)$ .
- Negative eigenvalues mean return toward equilibrium.
- Positive eigenvalues mean movement away from equilibrium.
- The report finds both signs, so it interprets the microstructure as a saddle-like system.

$$A = \begin{bmatrix} -\gamma & -\beta \\ \alpha & -\delta \end{bmatrix}$$

**Variables:** OBI = order book imbalance; Velocity\_y = mid-price speed;  $\alpha$  = OBI impact on price velocity;  $\beta$  = velocity feedback into OBI;  $\gamma$  = OBI mean-reversion rate;  $\delta$  = velocity damping rate;  $t$  = time.

# What the phase graph tells us

Most moments are calm, but extreme imbalance creates risk of directional motion.



OBI on x-axis, price velocity on y-axis

- The dense horizontal band at  $y \approx 0$  means price usually stays stable even when OBI changes.
- At OBI near +1, upward velocity spikes appear; buy pressure can push price up.
- At OBI near -1, downward velocity spikes appear; sell pressure can pull price down.
- So the main presentation message is: extreme imbalance is more informative than small imbalance.

Careful wording: this image is a phase-space scatter plot. The saddle-point claim comes from the fitted Jacobian/eigenvalues, not from the scatter alone.

**Variables:** OBI = x-axis imbalance; Velocity\_y = y-axis mid-price speed; points near zero show calm periods; extreme  $\pm$ OBI shows directional pressure.



# Logistic growth: liquidity returns with a limit

The order book refills after a shock, but not infinitely.

$$dV/dt = r V(t) \cdot (1 - V(t)/K)$$

- $V(t)$ : liquidity/volume available in the book at time  $t$ .
- $r$ : recovery rate — how fast liquidity comes back.
- $K$ : carrying capacity — the maximum sustainable liquidity level.
- The factor  $(1 - V/K)$  slows the growth as liquidity gets close to  $K$ .

Financial intuition: after a large trade or cancellation wave, market makers refill liquidity. They act fast, but they have capital limits and risk limits, so recovery must saturate.

small  $V \rightarrow$  fast growth  
 $V$  close to  $K \rightarrow$  growth  $\approx 0$

This creates the famous S-shaped recovery curve.

**Variables:**  $V(t)$  = liquidity/volume at time  $t$ ;  $dV/dt$  = rate of liquidity change;  $r$  = liquidity recovery speed;  $K$  = maximum book capacity/carrying capacity.

# Where the logistic solution comes from

Start from growth, add a maximum limit, then integrate.

1) Unlimited growth:  $dV/dt = rV$

2) Add remaining capacity:  $dV/dt = rV(1 - V/K)$

3) Integrate:  $\int dV/[V(1 - V/K)] = \int r dt$

4) Final S-curve:

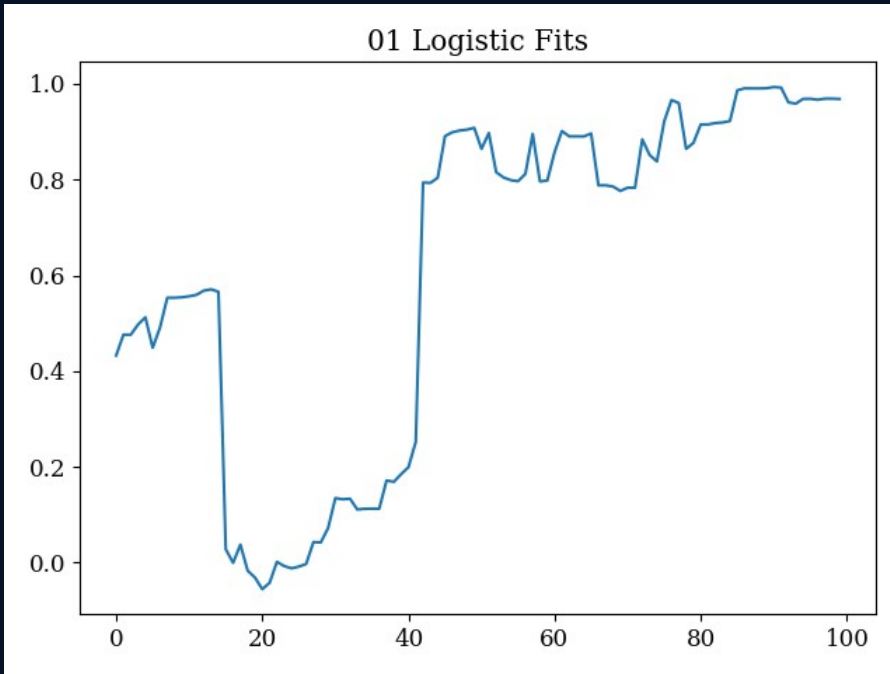
$$V(t) = K / [1 + ((K - V_0)/V_0)e^{(-rt)}]$$

- $V_0$  is the initial liquidity just after the shock.
- As  $t$  increases,  $e^{(-rt)}$  shrinks.
- So the denominator moves toward 1 and  $V(t)$  approaches  $K$ .
- That is exactly “liquidity recovery until capacity.”

**Variables:**  $V(t)$  = liquidity at time  $t$ ;  $V_0$  = starting liquidity after the shock;  $r$  = recovery speed;  $K$  = maximum liquidity capacity;  $C$  = integration constant.

# Logistic fit graphs: shock, refill, saturation

The fitted recovery speed  $r$  tells us how fast liquidity returns.



Example recovery after liquidity shock

- Left graph: liquidity drops sharply, then recovers toward a high stable level.
- The project fits 64 localized shock events using nonlinear least squares.
- Average fitted recovery rate:  $r = 18.32$ , meaning extremely fast replenishment.
- Trading interpretation: wait until liquidity crosses the logistic inflection point  $K/2$  before aggressive execution.

**Variables:** Left plot:  $V(t)$  recovery after shock; right plot: distribution of fitted recovery behavior;  $r$  = refill speed;  $K$  = capacity.

# Wronskian: checking if two signals are truly different

A model should not use duplicate features.

- In trading models, price, spread, volume, imbalance, and depth are used as features.
- If two features are basically the same signal, the model becomes redundant and unstable.
- The Wronskian is a calculus-based test for functional independence.
- The project checks whether price  $P(t)$  and spread  $S(t)$  carry independent information.

$$W(f_1, \dots, f_n)(t) = \det[ f_i^{(j-1)}(t) ]$$

Rule used here:

If  $W(t_0) \neq 0$  at at least one point, the functions are linearly independent.

Simple meaning: “they are not just copies of each other.”

# Wronskian formula used for price and spread

For two signals, the determinant has a simple form.

$$W(t) = \begin{vmatrix} P(t) & S(t) \\ P'(t) & S'(t) \end{vmatrix}$$

$$W(t) = P(t)S'(t) - S(t)P'(t)$$

- $P'(t)$  tells how price is changing.
- $S'(t)$  tells how spread is changing.
- If  $W(t)$  stays exactly zero, price and spread move like dependent functions.
- If  $W(t)$  fluctuates away from zero, price and spread contain different information.

Example: price rises while spread narrows. Then price and spread are clearly describing different market effects, so  $W(t)$  is non-zero.

Project result:

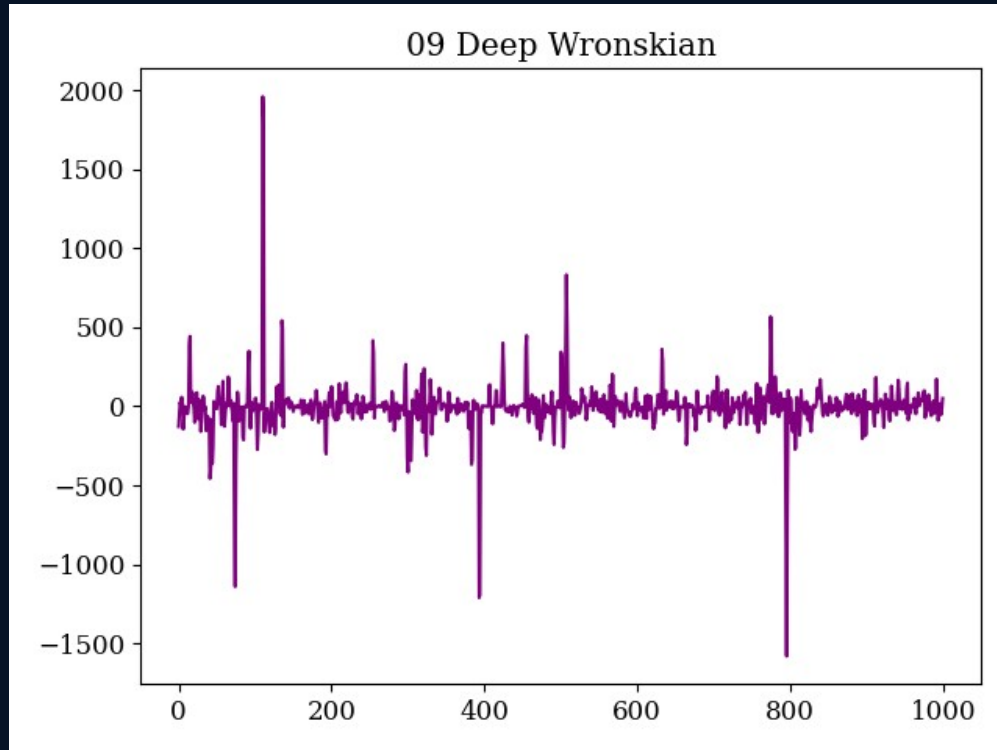
$$|W_{\text{avg}}| = 2.5902$$

$$\text{Std. deviation} = 211.59$$

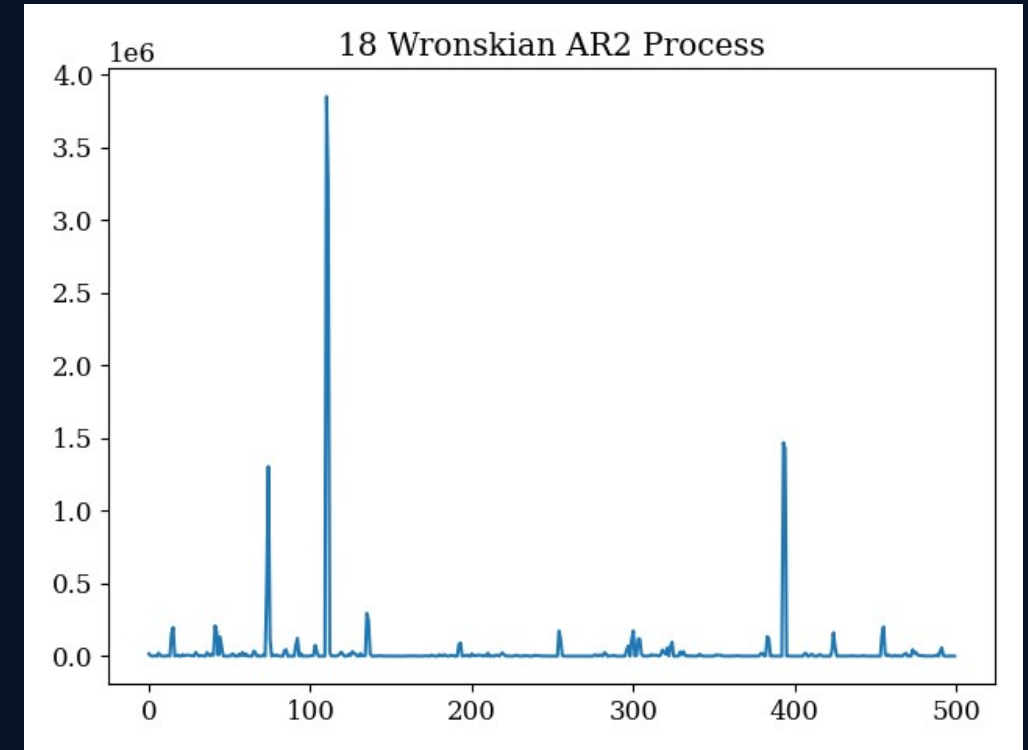
**Variables:**  $P(t)$  = price/mid-price;  $S(t)$  = bid-ask spread;  $P'(t)$ ,  $S'(t)$  = time derivatives;  $W(t)$  = Wronskian value measuring signal independence.

# Wronskian graphs: independence plus shock bursts

The signal is mostly around zero but repeatedly moves away from zero.



Deep Wronskian  $W(t) = P(t)S'(t) - S(t)P'(t)$



AR(2) spectral behaviour of the Wronskian process

- Left:  $W(t)$  never settles identically at zero, so price and spread are not redundant.
- Large spikes mark moments where the relation between price and spread changes sharply.
- Right: the AR(2)/spectral view shows rare bursts of very high energy.
- Trading use: Wronskian spikes can act as warnings of structural signal stress.

**Variables:**  $W(t)$  = Wronskian from price/spread; large spikes = moments where price and spread derivatives diverge sharply.

# 20-dimensional homogeneous linear ODE system

Now the whole 10-bid + 10-ask book is modeled together.

$$\dot{\mathbf{x}}(t) = \mathbf{A} \mathbf{x}(t)$$

- $\mathbf{x}(t)$  is a 20-dimensional vector of order-book depths.
- The first 10 components represent bid-side levels.
- The next 10 components represent ask-side levels.
- $\mathbf{A}$  is a  $20 \times 20$  interaction matrix: each entry says how one level affects another.

Why this matters: Level 1 does not move alone. A shock at one depth level can ripple across near and far liquidity.

$$\mathbf{x}(t) = c_1 e^{\lambda_1 t} \mathbf{v}_1 + \dots + c_{20} e^{\lambda_{20} t} \mathbf{v}_{20}$$

Each eigenvalue  $\lambda$  describes one independent mode of book movement.

**Variables:**  $\mathbf{x}(t)$  = 20-dimensional LOB depth vector;  $\dot{\mathbf{x}}(t)$  = derivative/change of that vector;  $\mathbf{A}$  = interaction matrix between bid/ask levels;  $\lambda_i$  = eigenvalue;  $\mathbf{v}_i$  = eigenvector;  $c_i$  = constants fixed by initial state.

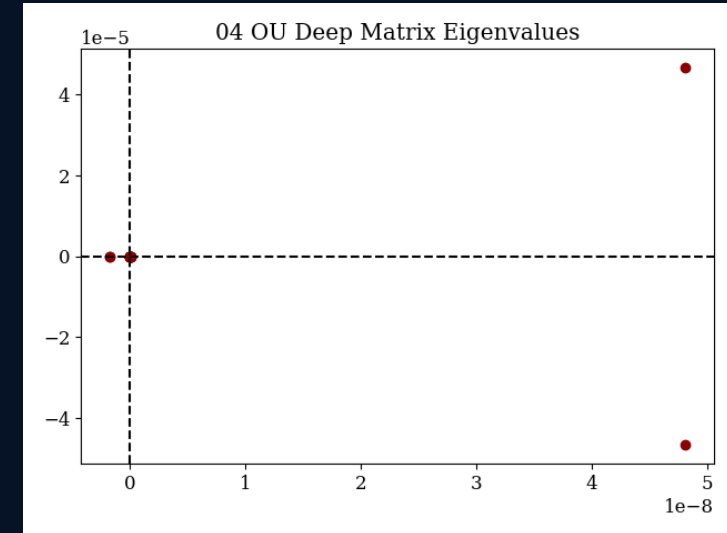
# Eigenvalues tell us stability vs breakout

The sign of the real part decides whether a mode decays or grows.

$\text{Re}(\lambda) < 0 \rightarrow$  disturbance decays  $\rightarrow$  stable / mean-reverting

$\text{Re}(\lambda) > 0 \rightarrow$  disturbance grows  $\rightarrow$  unstable / drifting

- The report finds negative eigenvalues near  $-0.0003$ : these suppress top-of-book deviations.
- It also finds positive eigenvalues near  $+0.0032$ : these allow deep-book pressures to grow.
- Because both signs appear, the system is saddle-like: some directions return to balance while others move away.
- Financial meaning: top levels may look stable, but deeper order blocks can still create momentum.



Eigenvalue plot of A

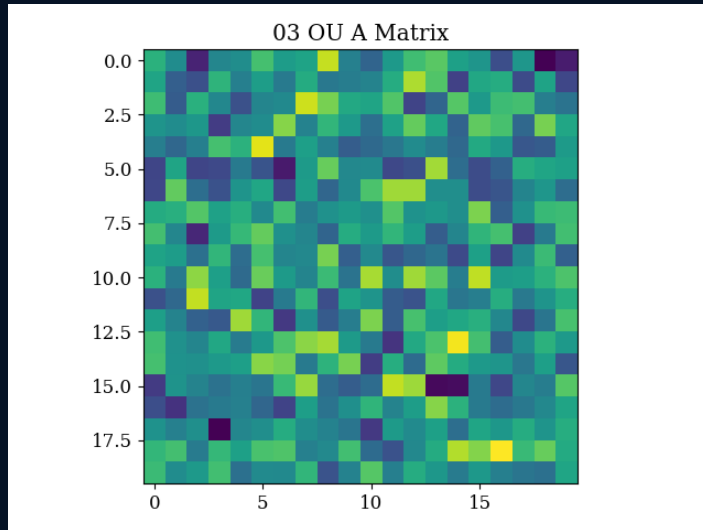
In presentation: "Eigenvalues are the stability DNA of the order book."

**Variables:**  $\lambda$  = eigenvalue of matrix A;  $\text{Re}(\lambda)$  = real part of  $\lambda$ ; negative  $\text{Re}(\lambda)$  means decay/mean reversion; positive  $\text{Re}(\lambda)$  means growth/unstable drift.

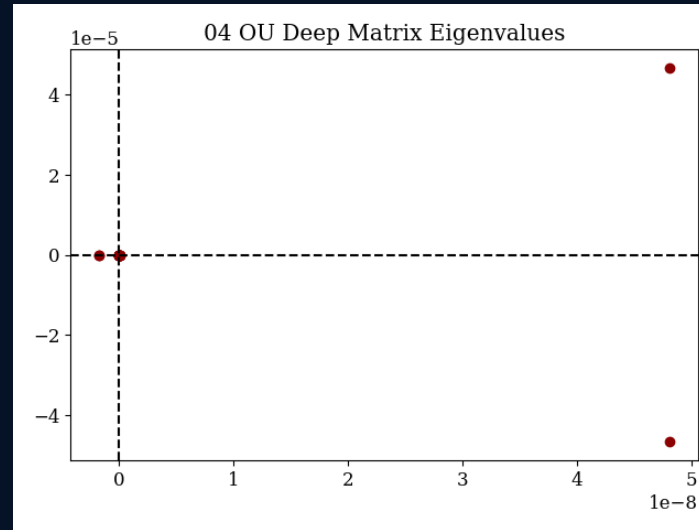


# Matrix, eigenvalues, and impulse response

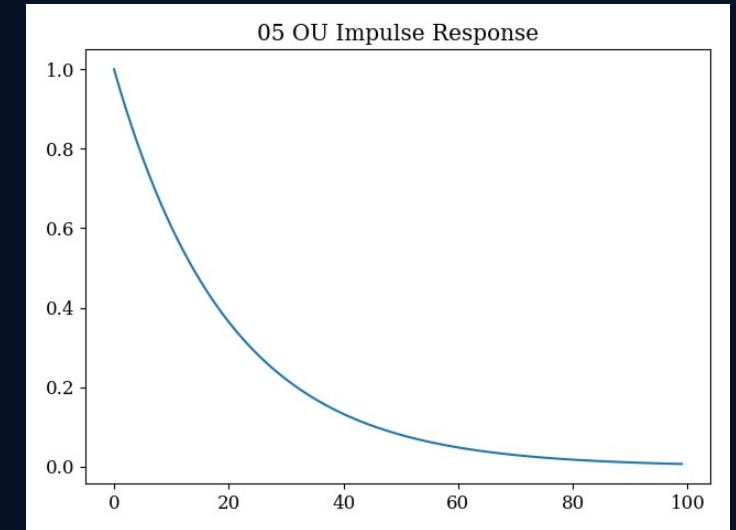
These three graphs show coupling, stability, and recovery speed.



A matrix heat map



Eigenvalues



Impulse response

- Heat map: each square is how one book tier reacts to another tier.
- Eigenvalue plot: positive and negative modes coexist, giving saddle-like behaviour.
- Impulse response: a stable shock decays back toward zero over time.
- Together: local liquidity can be absorbed, but some deep-book modes can still drive trends.

**Variables:** A = depth interaction matrix;  $\lambda$  = eigenvalue; impulse response = how a depth shock decays over time.

# Picard's method: solving an ODE by repeated guessing

Turn the differential equation into an integral equation, then refine repeatedly.

$$y' = f(t,y), \quad y(t_0)=y_0$$

$$y(t) = y_0 + \int [t_0 \text{ to } t] f(s, y(s)) \, ds$$

Analogy: it is like repeatedly correcting a forecast until the forecast becomes stable.

- Start with a simple first guess:  $\varphi_0(t) = y_0$ .
- Plug  $\varphi_0$  into the integral to get  $\varphi_1$ .
- Then plug  $\varphi_1$  into the integral to get  $\varphi_2$ .
- Keep repeating until the curves stop changing.

In the project, the function  $f(t,y)$  comes from empirical market flow, and  $y(t)$  represents short-horizon mid-price movement.

**Variables:**  $y(t)$  = unknown price path;  $y'$  = rate of price change;  $f(t,y)$  = market force from data;  $t_0$  = initial time;  $y_0$  = initial value.

# Picard recursion in the project

A transparent alternative to black-box short-horizon forecasting.

$$\varphi_0(t) = y_0$$

$$\varphi_1(t) = y_0 + \int f(s, \varphi_0(s)) \, ds$$

$$\varphi_{n+1}(t) = y_0 + \int f(s, \varphi_n(s)) \, ds$$

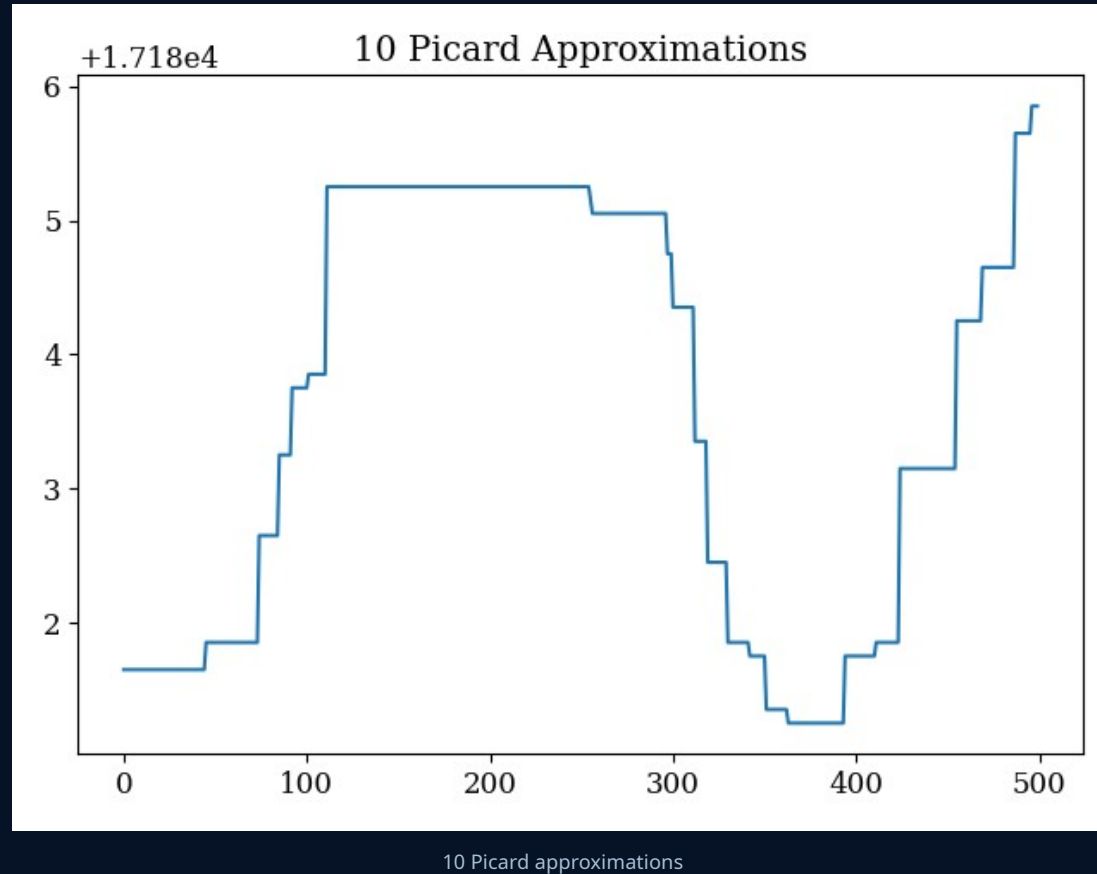
- Each iteration uses the previous curve to create a better curve.
- If the market is locally smooth, the iterations converge.
- If a regime shock breaks smoothness, convergence fails or error explodes.
- That failure becomes a warning signal for unpredictable conditions.

Project interpretation: classical ODE recursion behaves like an explainable unrolled sequence model.

**Variables:**  $\varphi_n(t)$  = nth Picard approximation;  $\varphi_0(t)$  = first guess;  $y_0$  = starting value;  $f(s, \varphi_n(s))$  = market force evaluated using current approximation;  $s$  = integration time variable.

# Picard graph: convergence of the approximation

The project simulates repeated Picard integrations on a volatile window.



- The curve shows successive approximations across a short price window.
- The report says 12 integrations reduce relative error from about 6.5 to 0.074.
- When the error shrinks, the ODE path is locally predictable.
- If the error stops shrinking, the market may have entered a shock regime.

Presentation line: "Picard gives both a forecast and a warning: convergence means stable local dynamics; divergence means the model should not trust the path."

**Variables:** Curve shows Picard approximation behaviour over iterations; lower residual means predicted path is stabilizing.

# Cauchy–Euler equations: power-law market memory

Some market shocks fade slowly, not exponentially.

- A simple exponential model says shocks disappear quickly at a fixed decay rate.
- But financial markets often remember shocks for longer than that.
- Cauchy–Euler ODEs naturally produce power-law solutions of the form  $y(t) = t^r$ .
- This is useful for spread shocks, volatility shocks, and post-event liquidity gaps.

exponential decay:  $Ce^{-kt}$

power-law decay:  $Ct^{-\mu}$

Power-law decay is slower in the tail. That means a shock can keep affecting spreads and volatility much longer than a simple exponential model predicts.

# How the report turns formulas into graphs

This section explains the implementation pipeline used on the order-book data.

- 1 Raw snapshots** Bid/ask prices and volumes are read at every timestamp.
- 2 Variables** Compute mid-price, spread, OBI, depth vectors, and liquidity volume.
- 3 Derivatives** Use finite differences: value change divided by  $\Delta t$ .
- 4 Fit / calculate** Apply the ODE formula: fit parameters, compute Wronskians, eigenvalues, or iterations.
- 5 Plot** Graph the resulting phase points, fitted curves, spectra, impulse responses, or residual errors.

$$\text{derivative} \approx (\text{next value} - \text{current value}) / \Delta t$$

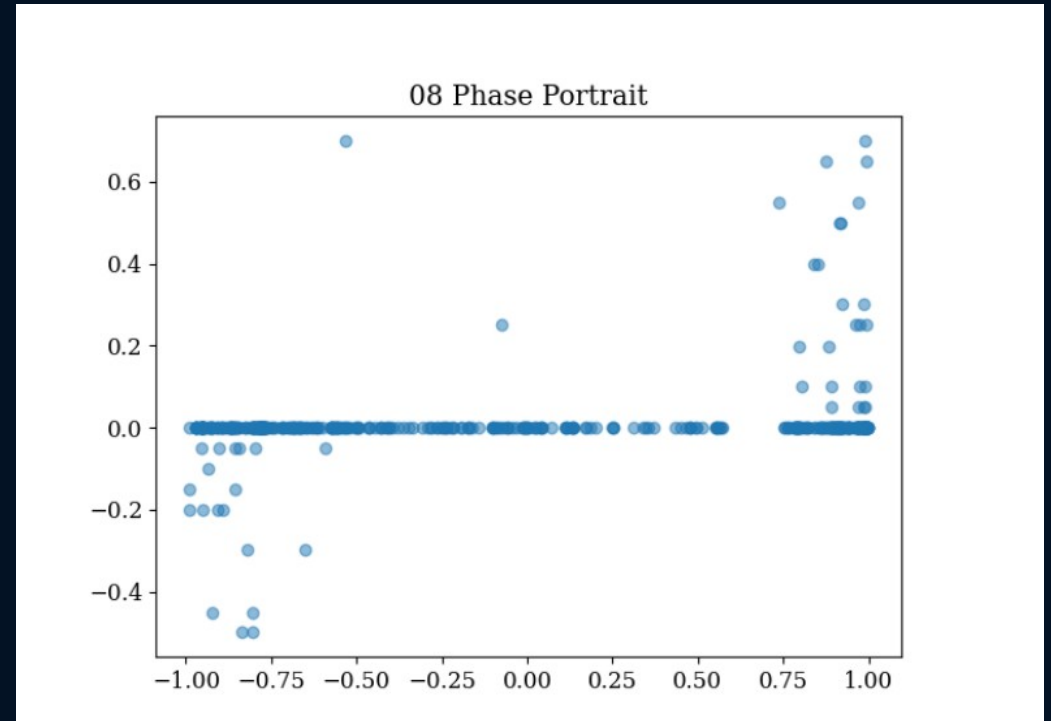
# Phase portrait: how the graph was created

The graph is not drawn directly from theory; it is drawn from computed OBI and velocity values.

$$\text{OBI}_t = (\text{BidVol}_t - \text{AskVol}_t) / (\text{BidVol}_t + \text{AskVol}_t)$$

$$\text{Velocity}_t \approx (\text{MidPrice}_{t+1} - \text{MidPrice}_t) / \Delta t$$

- For every timestamp, compute one point:  $(\text{OBI}_t, \text{Velocity}_t)$ .
- Plot all points together to see calm regions and breakout regions.
- The dense band around velocity = 0 means most ticks have little price movement.
- Extreme OBI values near  $\pm 1$  create the visible velocity spikes.



Generated from OBI and mid-price velocity

**Formula → Data use: each row becomes one coordinate in phase space.**

**Variables:** BidVol<sub>t</sub> = bid-side volume at time t; AskVol<sub>t</sub> = ask-side volume; OBI<sub>t</sub> = imbalance point; MidPrice<sub>t</sub> = average of best bid and ask; Δt = time gap between snapshots; Velocity<sub>t</sub> = mid-price change per unit time.

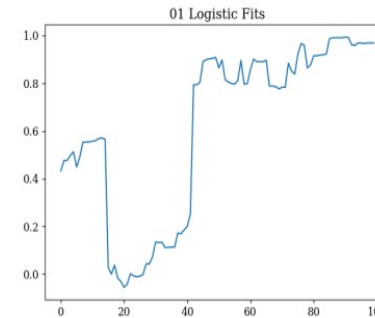
# Logistic recovery: how the fit graph was created

The formula was fitted to liquidity after shocks.

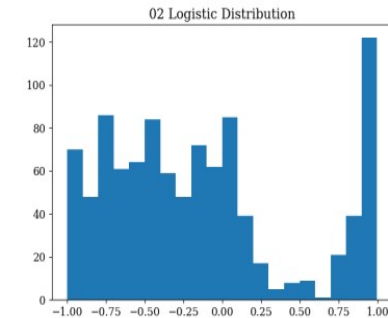
$$V(t) = K / \{1 + [(K - V_0)/V_0] e^{(-rt)}\}$$

- Find shock windows where liquidity suddenly drops.
- For each window, measure total available book volume  $V(t)$  after the drop.
- Fit the logistic curve to the recovery using non-linear least squares.
- The fitted parameter  $r$  becomes the liquidity recovery speed.
- The report's fitted mean is  $r = 18.32$ , meaning very fast recovery.

by  $r$ , the firm's localized order sizing thresholds.



(a) Geometric logistic fits against isolated empirical liquidity sweeps, matching the classic  $S$ -curve formulation.



(b) Numerical distribution array revealing the true market-making recovery constraint parameter  $r$ .

Figure 2: Continuous Population modeling mapped completely to limit order execution architectures.

The GPU evaluation isolated 64 severe localized shock instances. Applying the `scipy.optimize` non-linear least squares fit directly mapped the data to  $r = 18.32$  as the structural mean. Meaning that true organic replenishment possesses a functional half-life velocity near  $t = 0.03$  to  $0.05$  seconds before the maximum carrying capacity asymptotically caps the algorithmic accumulation. A quantitative firm mathematically exploits this by holding its aggressive market-order entries precisely until the logistic curve passes the inflection point  $\frac{K}{2}$ .

Shock → refill → saturation

**fastest refill happens near  $V = K/2$**

**Variables:**  $V(t)$  = post-shock liquidity;  $V_0$  = liquidity at start of recovery;  $K$  = maximum recoverable liquidity;  $r$  = recovery speed;  $t$  = time after shock.



# Wronskian: how independence was computed

Price and spread were tested as two separate signals.

$$W_t = P_t S'_t - S_t P'_t$$

- Compute mid-price  $P_t$  and spread  $S_t$  for every tick.
- Approximate  $P'_t$  and  $S'_t$  using central differences.
- Calculate  $W_t$  at every timestamp.
- Plot  $W_t$  over time: if it never collapses to zero, the features are not redundant.
- This produced the “Deep Wronskian” plot.

## 4 3. Wronskian Determinants and Linear Independence

### 4.1 Rigorous Mathematical Formulation

When processing multi-variate predictor variables (features) for systematic trading rules, a critical mathematical breakdown occurs if the signals share perfect collinearity. To conclusively prove functional non-redundancy along continuous manifolds, we deploy the Wronskian test.

**Definition 4.1.** Let  $f_1, f_2, \dots, f_n$  be  $(n-1)$  times differentiable functions operating dynamically on a continuous temporal interval  $I$ . Their geometric Wronskian  $W(f_1, \dots, f_n)(t)$  is firmly established as:

$$W(t) = \begin{vmatrix} f_1(t) & f_2(t) & \dots & f_n(t) \\ f'_1(t) & f'_2(t) & \dots & f'_n(t) \\ \vdots & \vdots & \ddots & \vdots \\ f_1^{(n-1)}(t) & f_2^{(n-1)}(t) & \dots & f_n^{(n-1)}(t) \end{vmatrix} \quad (11)$$

**Theorem 4.1** (Non-vanishing Theorem). If  $W(t_0) \neq 0$  efficiently for at least one critical  $t_0 \in I$ , the functional solutions  $\{f_1, \dots, f_n\}$  are strictly, perfectly linearly independent across  $I$ .

### 4.2 Feature Defensibility in Crypto Pair Models

In trading, the  $f_1$  vector might be the continuous Price  $P(t)$ , whilst  $f_2$  is the ongoing bid-ask Spread dimension  $S(t)$ . We utilized central-difference numerical derivation heavily across PyTorch CUDA blocks to establish the base calculus mappings  $P'(t)$  and  $S'(t)$ , mapping the subsequent determinant  $W = P(t)S'(t) - S(t)P'(t)$  continuously spanning 3,730,870 micro units.

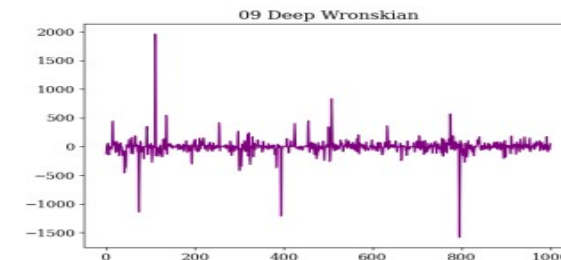


Figure 3: Continuous fluctuations showcasing the dense localized Wronskian determinant oscillating.  $W$  never stabilizes mechanically at identically zero.

The mean amplitude of the Wronskian resides at  $|W_{avg}| = 2.5902$ , maintaining massive spatial separation from 0 alongside a high standard deviation of 211.59. Consequently,  $P$  and  $S$  absolutely represent uncorrelated signal vectors under an ODE paradigm.

Wronskian fluctuations over time

**Interpretation: price and spread carry different information.**

**Variables:**  $P_t$  = price/mid-price at time  $t$ ;  $S_t$  = spread at time  $t$ ;  $P'_t$  and  $S'_t$  = finite-difference derivatives;  $W_t$  = computed Wronskian signal.

# Wronskian AR(2) spectrum: how the spike graph was produced

After computing  $W_t$ , the project studies whether its shock bursts have structure.

$$W_t \approx a_1 W_{t-1} + a_2 W_{t-2} + \varepsilon_t$$

- Use the Wronskian time series from the previous slide.
- Fit an AR(2) model using two previous  $W$  values.
- Convert the fitted process into a spectral/power view.
- Tall spikes represent rare frequencies/lags where the Wronskian energy becomes very large.
- This supports the idea of mostly stable behaviour with occasional microstructure shocks.

Raj Lohar (BSD-CC-2414)

Dynamical Systems in Crypto Microstructure

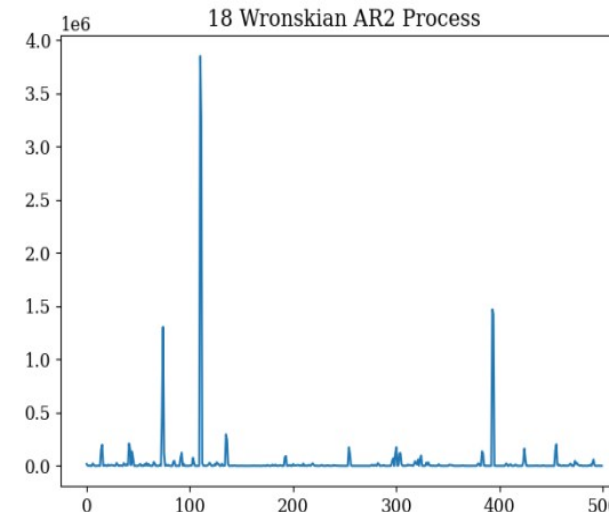


Figure 4: Examining the Autoregressive spectral behavior of the LOB's Wronskian operator, validating that structural signal breakdowns possess massive stationarity constraints.

Spectral spikes in the Wronskian operator

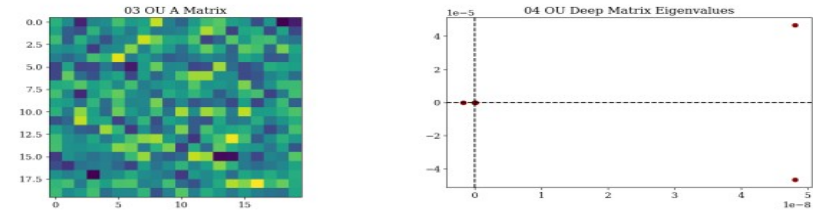
**Variables:**  $W_t$  = Wronskian at time  $t$ ;  $W_{t-1}, W_{t-2}$  = previous two Wronskian values;  $a_1, a_2$  = AR(2) coefficients;  $\varepsilon_t$  = residual/noise shock.

# 20-level linear ODE: how matrix and eigenvalue graphs were created

$$\mathbf{x}(t) = [\text{BidVol}_1 \dots \text{BidVol}_{10}, \text{AskVol}_1 \dots \text{AskVol}_{10}]^T$$

$$\dot{\mathbf{x}}(t) \approx [\mathbf{x}(t+\Delta t) - \mathbf{x}(t)] / \Delta t \approx \mathbf{A} \mathbf{x}(t)$$

- Build  $\mathbf{x}(t)$  from 10 bid volumes and 10 ask volumes.
- Calculate  $\dot{\mathbf{x}}(t)$ , the change in the depth vector.
- Fit matrix  $\mathbf{A}$  so that  $\mathbf{A} \mathbf{x}(t)$  best explains  $\dot{\mathbf{x}}(t)$ .
- Plot  $\mathbf{A}$  as a heat map to show coupling between levels.
- Compute eigenvalues of  $\mathbf{A}$  to separate stable and unstable modes.



(a) Empirical Matrix  $\mathbf{A}$  heat map outlining the deep dimensional interaction between near and far liquidities. (b) Plot of the pure 20 eigenvalue scalars. The existence of both fractional positive and deep negative dimensions confirms Saddle Point instability.

Figure 5: LOB flow coupling visualization and rigorous algebraic spectral decay.

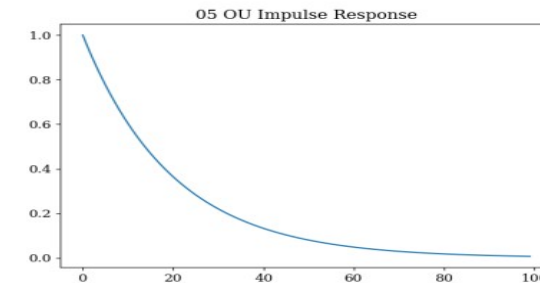


Figure 6: Visualizing the resulting continuous multi-variable homogeneous impulse responses; simulating a shock injected dynamically against  $\mathbf{x}(0)$ , followed rigorously by complex uniform decay over a massive interval.

A matrix, eigenvalues, and impulse response

**Negative eigenvalue → decay; positive eigenvalue → drift/breakout.**

**Variables:**  $\mathbf{x}(t)$  = vector of 10 bid volumes and 10 ask volumes;  $\dot{\mathbf{x}}(t)$  = change in depth vector;  $\mathbf{A}$  = fitted 20×20 interaction matrix;  $\Delta t$  = snapshot time gap.

# Picard approximation: how the convergence graph was created

$$\varphi_{n+1}(t) = y_0 + \int f(s, \varphi_n(s)) ds$$

- Choose a volatile 100-step price window.
- Start with a simple initial path:  $\varphi_0(t)=y_0$ .
- Use the empirical order-flow force  $f(t,y)$  to integrate a better path.
- Repeat the integral update 12 times.
- Plot approximation/residual values to show convergence.

## 6.2 Trajectory Projection Under Imbalance Pressures

We utilized Picard Integrals deeply on our tensor infrastructure to iteratively map continuous crypto futures approximations spanning a highly volatile 100-step subset, effectively asking: can localized differential mapping recursively forecast true price mechanics dynamically without a neural network model? We integrated Mid-Price movements explicitly mapping empirical flow as the physical continuous forcing field  $f(t,y)$ .

The convergence visualizes the power of classical ODE iterative theories acting identically to deep recursive neural frameworks (unrolled RNN architectures process structurally highly similar sequence operators). Mathematical convergence explicitly proves the limits of predictability: whenever a massive regime shifting external shock breaks the Lipschitz metric bound dynamically, the recursion shatters visually, informing the quant trader to rapidly pull inventory structures.

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Raj Lohar (BSD-CC-2414)

Dynamical Systems in Crypto Microstructure

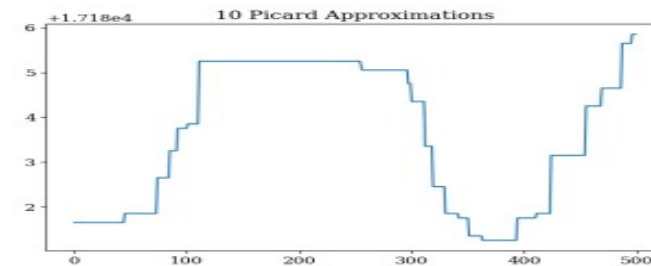


Figure 7: Simulating 12 distinct Picard successive integrations. The error relative differential drastically and efficiently tightens from the massive 6.5 initial start boundary precisely downwards collapsing into 0.074 deviation space mappings.

Successive Picard approximations

**Data result: residual error shrinks from about 6.5 to 0.074.**

**Variables:**  $\varphi_n(t)$  = nth predicted price path;  $y_0$  = initial price;  $f$  = empirical order-flow force;  $s$  = dummy integration variable;  $n$  = iteration number.

# Complete formula-to-data mapping

Each graph came from turning raw LOB columns into a mathematical object.

| Topic             | Data input                       | Graph output                             |
|-------------------|----------------------------------|------------------------------------------|
| Phase portrait    | OBI, MidPrice velocity           | scatter of $(OBI_t, Velocity_t)$         |
| Logistic recovery | liquidity after shock            | fit $V(t)$ and estimate $r$              |
| Wronskian         | price and spread derivatives     | plot $W_t = P_t S'_t - S_t P'_t$         |
| 20-level ODE      | 10 bid + 10 ask volumes          | fit $A$ , plot heat map/eigenvalues      |
| Picard            | short price window + force field | iterate $\varphi_n$ and plot convergence |
| Cauchy–Euler      | post-shock spread/volatility     | fit power-law $t^{(-r)}$                 |

**Main takeaway: every graph is a data-transformed version of an ODE concept, not a separate unrelated plot.**

# Key empirical findings from the project

These are the numbers to remember from the analysis.

**−0.0003 to  
+0.0032**  
stability spectrum

**$r = 18.32$**   
liquidity recovery

**0.074**  
Picard residual

**$r \approx 0.94$**   
memory exponent

**2.5902**  
Wronskian avg

- Saddle-like behaviour: the book has stable and unstable directions at the same time.
- Liquidity replenishment is very fast after shocks, but capacity-limited.
- Price and spread are not redundant signals under the Wronskian test.
- Shock memory follows a slower power-law pattern, not just exponential decay.

# Trading implications in simple language

The math is translated into execution and risk ideas.

- Heal-rate execution: after a liquidity sweep, wait about 40–80 ms spacing so the book can refill.
- Lead-lag imbalance: strong OBI can appear before mid-price movement, so imbalance becomes a short-horizon signal.
- Deep-tier momentum: do not rely only on level-1 noise; deep-book pressure can drive breakout moves.
- Power-law tail risk: after a shock, keep spreads wider longer because memory fades slowly.

Important: these are research interpretations, not guaranteed trading profits. The model explains mechanisms, but real markets also include stochastic shocks, fees, latency, and execution constraints.

# How the ODE tools map to an algorithmic architecture

Each mathematical module can become a signal or risk-control component.

- Phase portrait/eigenvalue module: detect whether the book is in stable reversion or unstable drift.
- Logistic module: estimate post-shock liquidity recovery and decide when to resume execution.
- Wronskian module: monitor whether price and spread remain independent or collapse into signal stress.
- Cauchy–Euler module: estimate how long shock memory should affect quoting and risk limits.
- Picard module: decide whether short-term trajectory approximations are currently stable.



# Limitations and careful wording

This slide helps you answer questions honestly during presentation.

- The original data is discrete; ODE modeling is a continuous approximation justified by very high tick density.
- Some parameters such as  $\alpha$ ,  $\beta$ ,  $\gamma$ ,  $\delta$  are symbolic in the report; the combined fitted Jacobian/eigenvalues are emphasized more than individual parameter estimates.
- Graphs often use normalized or transformed axes, so they should be explained by behaviour, not only raw units.
- Deterministic ODE tools reveal structure, but they do not remove randomness, news shocks, latency, or execution costs.
- The project should be presented as a mathematical framework for understanding microstructure, not as a finished live-trading system.

Best answer if questioned: “The project shows how ODE methods can organize and explain high-frequency LOB behaviour; further validation would be needed before production trading.”

# Conclusion: what the project achieved

A continuous dynamical-systems lens for crypto order books.

- Converted high-frequency crypto LOB data into ODE-style state variables and derivatives.
- Used phase portraits and eigenvalues to study stability and saddle-like drift.
- Used logistic growth to explain bounded liquidity recovery after shocks.
- Used Wronskian determinants to validate signal independence between price and spread.
- Used Picard iteration and Cauchy–Euler memory modeling to explain short-term trajectory approximation and slow post-shock decay.

Final message: the order book is not only a table of orders — at high frequency it behaves like a dynamic physical system with stability, liquidity recovery, memory, and shock response.

# Thank You

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## Questions?

This presentation explains the project in simple language while preserving the mathematical structure and original graphs.

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